

Modeling 2D Marshak Waves

Understanding the physics of x-ray radiation driven supersonic heat waves

Yuval Kochavi

The Hebrew University of Jerusalem

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Photon Transport and Energy Exchange

Photon transport (Boltzmann equation):

$$\frac{1}{c} \frac{\partial I}{\partial t} + \hat{\Omega} \cdot \nabla I + (\sigma_a(T_m) + \sigma_s(T_m))I = \sigma_a(T_m)B(T_m) + \frac{\sigma_s(T_m)}{4\pi} \int_{4\pi} I d\hat{\Omega} + S(\hat{\Omega}, \vec{r}, t) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c} \frac{\partial T_m}{\partial t} = \sigma_a(T_m) \left(\frac{1}{c} \int_{4\pi} I d\hat{\Omega} - aT_m^4 \right) \quad (2)$$

Gray Coupled Equations and LTE

We get the "gray" coupled equations:

$$\begin{aligned}\frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) &= c\sigma(U - E), & U &= aT_m^4, & E &= aT_R^4 \\ \frac{\partial e}{\partial t} &= c\sigma(E - U), & e &= fT_m^\beta \rho^{-\mu}\end{aligned}$$

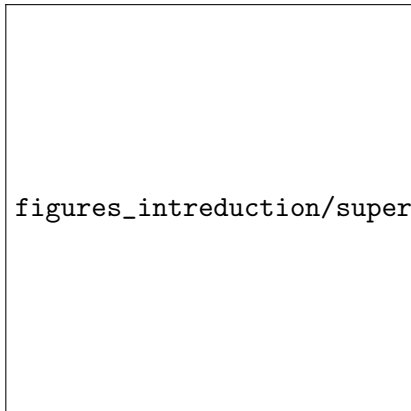
Finally, assuming complete thermodynamic equilibrium (CTE) between the radiation field and the material ($T_R = T_m$):

$$\frac{\partial T^4}{\partial t} + f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right) \quad (3)$$

Assuming $\rho f T^\beta \rho^{-\mu} \gg aT^4$ (True until around 3keV):

$$f\rho^{-\mu} \frac{\partial T^\beta}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right)$$

Motivation



figures_introduction/supersonic.png

- Given the material properties equations:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

- The diffusion coefficient D is not linear.

Marshak Waves Experiments

Using a Hohlraum as an x-ray generator, a supersonic heatwave propagates in the low-density foam.

figures_intredution/T_drive.png figures_intredution/setup.png

Marshak Waves Experiments

figures_introduction/method1.png

figures_introduction/flux.png



figures_introduction/method2.png



Experiments tested

The experiment	foam type	density (mg/cc)	temperature (eV)
Massen <i>et al.</i>	$C_{11}H_{16}Pb_{0.3852}$	80	100–150
Back <i>et al.</i>	SiO_2 , Ta_2O_5	10–50	85–190
Xu <i>et al.</i>	C_6H_{12} , $C_6H_{12}Cu_{0.394}$	50	160
Ji-Yan <i>et al.</i>	C_8H_8	160	175
Rosen and Keiter <i>et al.</i>	$C_{15}H_{20}O_6$, $C_{15}H_{20}O_6Au_{0.172}$	45–70	200–210
Moore <i>et al.</i>	SiO_2 , C_8H_7Cl	90–120	310
Courtois <i>et al.</i>	SiO_2	19 – 29	170

1.5D model

HR Model - Governing Equations

The "Black" diffusion equation (neglecting radiation energy density $\frac{\partial E}{\partial t} \rightarrow 0$):

$$\underbrace{\frac{\partial E}{\partial t}}_{\rightarrow 0} + \frac{\partial e}{\partial t} = \frac{\partial}{\partial x} \left(\frac{c}{3\rho\kappa} \frac{\partial(aT^4)}{\partial x} \right)$$

With specific energy and opacity models:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

where $f, \mu, g, \lambda, \alpha, \beta$ are material-dependent parameters.

HR solved the diffusion equation using perturbation theory and found some interesting results.

HR Model - Heat Front Solution

The heat front position $x_F(t)$ is given by:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

where:

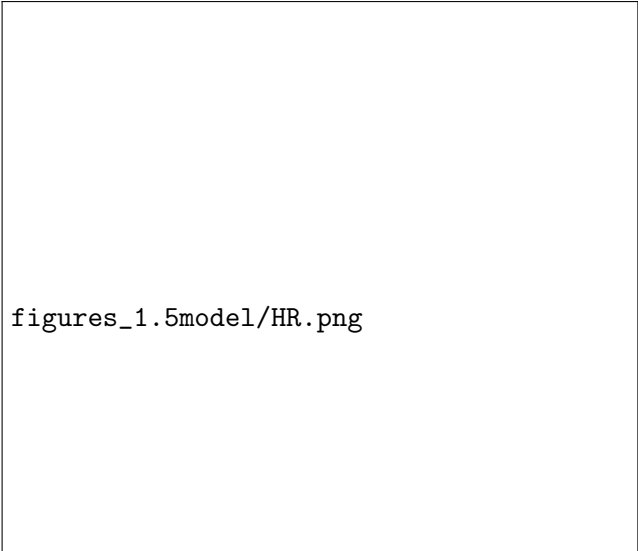
$$C = \frac{16g\sigma_{SB}}{3f(4 + \alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4 + \alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where T_s is the surface temperature at $x = 0$.

HR 1D Model Limitations

HR misses the real experimental results of Back et al. by a factor of 2!



figures_1.5model/HR.png

1.5D model

1D effects

Marshak Boundary Condition: $T_D \neq T_s$

- Assuming $T_s = T_D$ overestimates the wave speed.
- Instead, energy conservation at the boundary requires:

$$\sigma_{SB} T_D^4(t) = \sigma_{SB} T_s^4(t) + \frac{F(0, t)}{2}$$

where $F(0, t)$ is the net flux entering the foam.

Marshak Boundary Condition: $T_D \neq T_s$

Applying the Marshak boundary condition slows the propagation front significantly:



figures_1.5model/marshak.png

1.5D model

2D effects

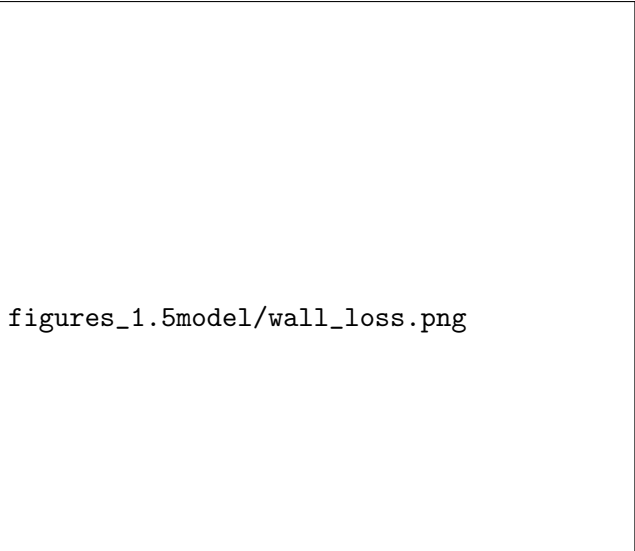
- Energy is lost from the foam into the tube walls.
- Can be modeled for different wall materials: **Gold**, **Copper**, **Beryllium**, and **Vacuum**.
- For **Gold**:

$$E_{\text{gold}}(t) = 0.59 T_s^{3.35} [t - t_0(x)]^{0.59} \quad [\text{hJ/mm}^2]$$

where $t_0(x)$ is the front arrival time at position x .

Energy Wall Loss

Including wall loss reduces the front velocity:

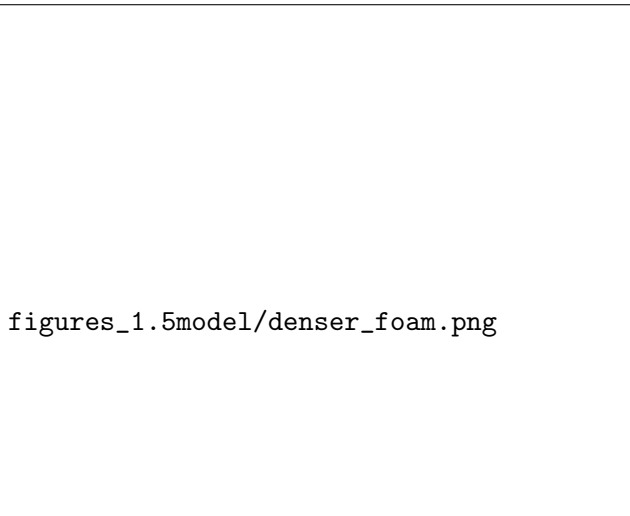


figures_1.5model/wall_loss.png

- High-Z wall material (e.g. Gold) absorbs energy and ablates inward into the foam.
- **Changing Foam Radius:** The inlet and tube radius shrink, reducing the heat wave cross-section.
- **Changing Foam Density:** The ablated wall material compresses and densifies the adjacent foam.
- Both effects must be incorporated to model the dynamic tube geometry.

Wall Ablation

Inward wall ablation compressively densifies the foam, which significantly slows down propagation:



Effective Mean Free Path

When $\lambda_R \geq 2R$, wall collisions control the transport instead of diffusion.
Can be modeled as:

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}$$

where λ_R is the Ross Mean Free Path.

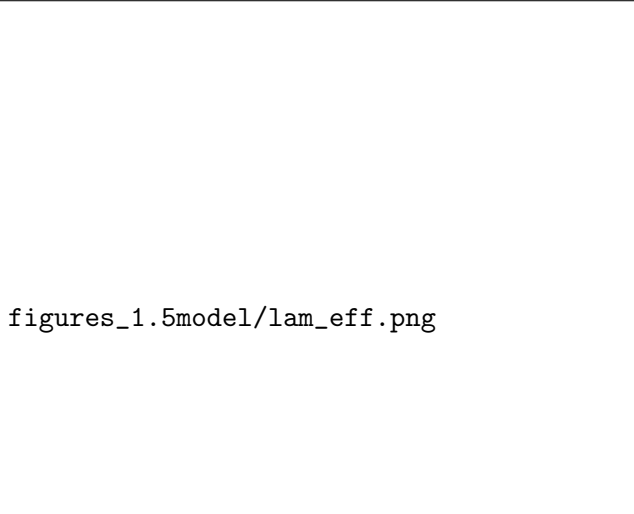
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = ((2R)^{-n} + \lambda_R^{-n})^{-1/n}$$

where n controls the "tightness" of the transition.

Effective Mean Free Path

Including the effective mean free path correction matches the simulation to the experimental data at early times:



figures_1.5model/lam_eff.png

Comparison to many experiments

figures_1.5model/C8H7Cl.png

Moore et al. JQSRT, (2015)

figures_1.5model/Ta2O5.png

Back et al. POP, (2000)

figures_1.5model/SiO2_low_energy.png

figures_1.5model/Ji-Yan.png

Full 2D model

Modeling the front curvature

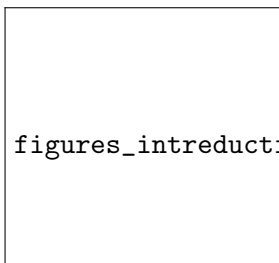
Understanding the front curvature

Hurricane & Hammer (2006) solved the laplace equation (assumed constant opacity and neglected e):

$$\underbrace{\frac{\partial e}{\partial t}}_{\rightarrow 0} = \frac{4}{3} \nabla \cdot \left(\frac{1}{\rho \kappa} \nabla (T^4) \right)$$

and got the 2D front in Cartesian coordinates (where $\varepsilon = \frac{3}{4} \rho \kappa L (1 - a)$):

$$x_F(y, t) = \frac{L}{\varepsilon} \cosh^{-1} \left(1 + \frac{D \varepsilon t}{L^2} \right) \cos \left(\frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}$$



figures_intreduction/hurricane.png

Cylinder Coordinates

The model was re-developed in cylindrical coordinates (*Courtois et al. POP, (2022)*) to match cylindrical geometries:

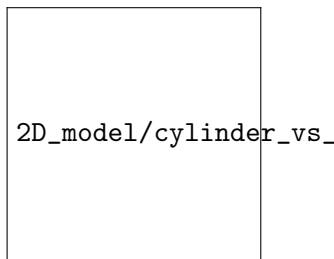
$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D\kappa_0^2 t}{2} \right) J_0(\kappa_0 r) + \text{HOT}$$

Planar $\longleftrightarrow$ **Cylindrical Analogy:**

$\cos(ky) \longleftrightarrow J_0(\kappa r)$ For small ε :

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4} \rho \kappa R (1 - a)$$

where a is the wall albedo.



Applying on our model: Dynamic Albedo

Until now, wall albedo was treated as a constant parameter. We update it dynamically in time:

$$\alpha(t) = \frac{F_{\text{out, wall}}}{F_{\text{in, wall}}} = \frac{\sigma_{SB} T_{s, \text{interface}}^4 - \frac{dE_{\text{wall}}}{dt} / 2}{\sigma_{SB} T_{s, \text{interface}}^4 + \frac{dE_{\text{wall}}}{dt} / 2}$$

This provides a much more physical description of the energy reflection.

where the curvature goes as

$J_0 \left(\frac{\sqrt{2\varepsilon(t)}}{R} r \right)$, and $\varepsilon(t)$ scales with $(1 - \alpha(t))$.

2D_model/albedo.png

**Problem: the longitudinal
front is way off!**

**Luckily, we have a direct
longitudinal position:**

$$x_F(r, t) = x_{1.5D}(t) \cdot J_0(\kappa_0(t)r)$$

Full 2D model

Comparing to 2D supersonic Diffusion Simulation

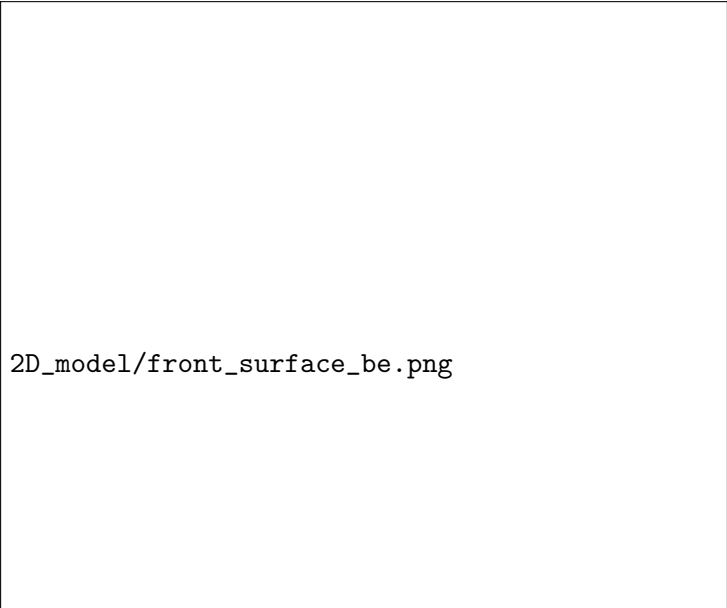
Gold coating - SiO₂

2D_model/front_surface_gold.png

100 times transparenter Gold coating - SiO₂

2D_model/front_surface_gold100.png

2D_model/front_surface_copper.png

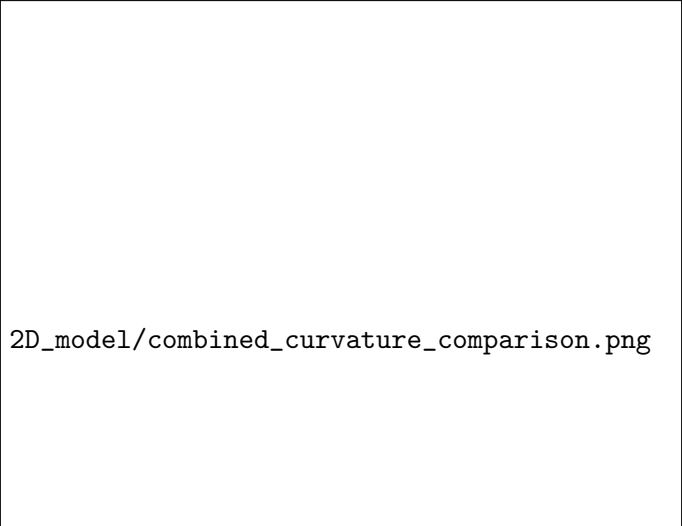


2D_model/front_surface_be.png

2D_model/front_surface_vacuum.png

Gold vs. Copper Coating Comparison

French experiment *Courtois et al. POP, (2024)* which tests the arrival time of the front as a function of r for both gold and copper coats:



2D_model/combined_curvature_comparison.png

Full 2D model

A full 2D Temperature distribution

A full 2D Temperature distribution

One can use the Heyney profile:

$$T(z, r, t) \approx T_S(t) \left[1 - \frac{z}{z_F(r, t)} \right]^{\frac{1}{4+\alpha-\beta}}$$

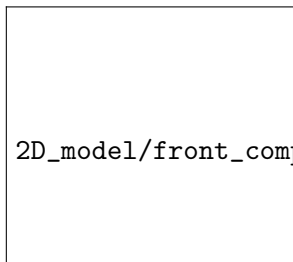
2D_model/2D_front_spatial.png

2D_model/temperature_heat

2D Temperature Distribution — Moore ($\text{C}_8\text{H}_7\text{Cl}$)

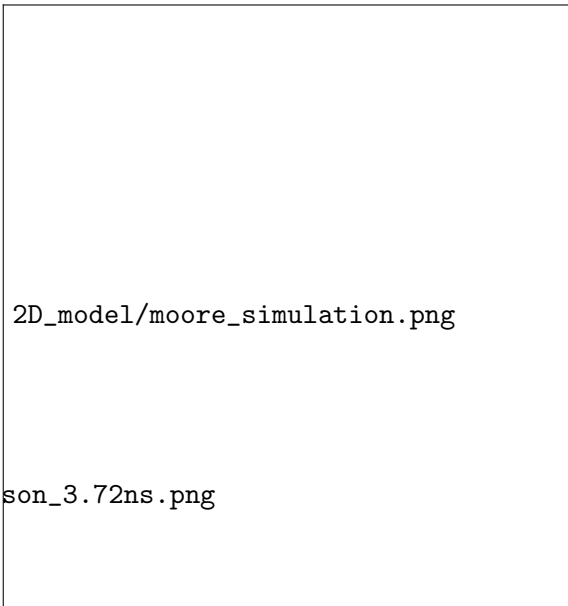
Side-by-side comparison of the 2D temperature distribution at $t = 3.72$ ns:

- **Left:** Diffusion simulation taken from *Moore et al. JQSRT, (2015)*.
- **Right:** Our full 2D model.



2D_model/front_comparison_3.72ns.png

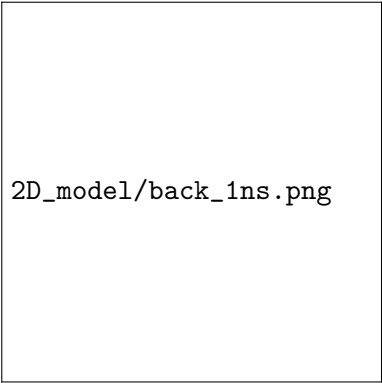
2D_model/moore_simulation.png



2D Temp. Distribution — Back SiO₂ (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP*, (2000) experiment.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.

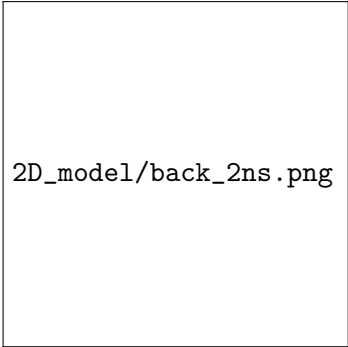


2D_model/back_1ns.png

2D Temp. Distribution — Back SiO₂ (2.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.0$ ns:

- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.

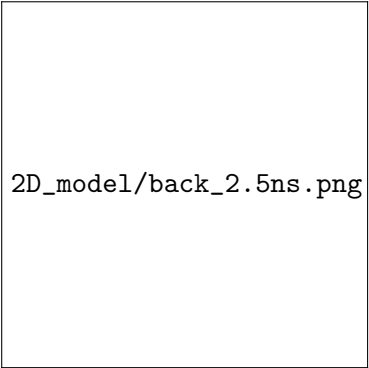


2D_model/back_2ns.png

2D Temp. Distribution — Back SiO₂ (2.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.5$ ns:

- **Left:** Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.

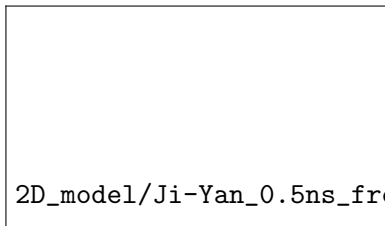


2D_model/back_2.5ns.png

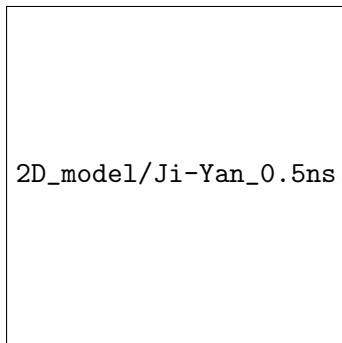
2D Temp. Distribution — Ji-Yan (0.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 0.5$ ns:

- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D_model/Ji-Yan_0.5ns_front.png

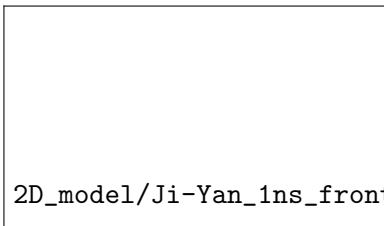


2D_model/Ji-Yan_0.5ns.png

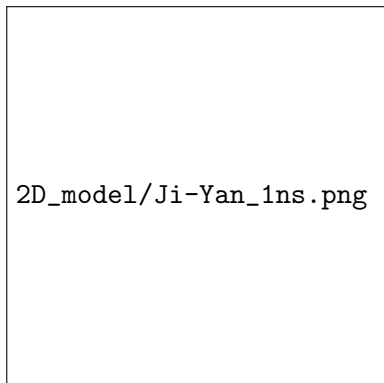
2D Temp. Distribution — Ji-Yan (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D_model/Ji-Yan_1ns_front.png

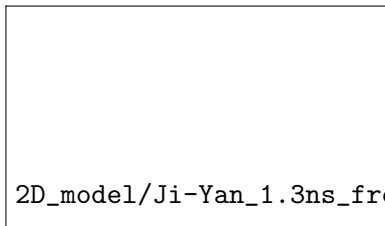


2D_model/Ji-Yan_1ns.png

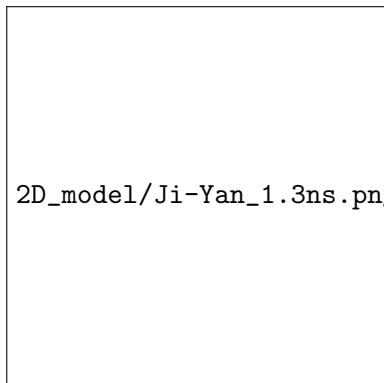
2D Temp. Distribution — Ji-Yan (1.3 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.3$ ns:

- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D_model/Ji-Yan_1.3ns_front.png



2D_model/Ji-Yan_1.3ns.png

Flux Curvature Comparison - Back SiO₂ PRL

2D_model/prl_T4.png

2D_model/prl.png



2D_model/back_prl_curve.png

Flux Curvature Comparison - Back Ta2O5 POP

2D_model/T4_spatial_heatmap_2.0ns.png



2D_model/flux_curvature_post_breako

2D_model/Ta205_intensity.png

- **Successful 2D Modeling:**

- We have successfully modeled 2D supersonic Marshak waves.
- Our unified model captures front curvature and temperature distribution.

- **Deep Physical Insight:**

- We can now understand the underlying physics of radiation transport and energy exchange in a very good, intuitive way.

Thank You!

Questions?